HKUST COMP 4421: Digital Image Processing

Assignment 1 Formal Solutions and Engineering Report

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Fall Semester 2026-27

Course Architecture & Official AI Policy Compliance:
In strict accordance with the HKUST COMP 4421 syllabus regulations (*"Students are free to use generative AI on their works, and should acknowledge it when used"*), generative AI models were employed in an auxiliary, consultative capacity for algebraic cross-verification, script boundary validation, and typesetting optimization. All mathematical derivations, algorithmic implementations, experimental code, and written responses were independently executed, audited, and mathematically verified by the author. A full statement is documented in Appendix B.

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1 Problem 1: Image Arithmetic and Histograms (10 pts)

1.1 Mathematical Framework

Let $\Omega = \{(x, y) \in \mathbb{Z}^2 \mid 0 \le x < M,\ 0 \le y < N\}$ denote the discrete spatial lattice of size $M = 40$ and $N = 50$, with total cardinality $|\Omega| = 2000$. Pixel intensities are quantized on the 5-bit discrete integer ring $\mathbb{Z}_{32} = \{0, 1, \ldots, 31\}$. Image A is a homogeneous field $A(x, y) \equiv 6$, and Image B is characterized by its empirical histogram $n_B(r) = \#\{(x, y) \in \Omega \mid B(x, y) = r\}$.

1.2 (a) Saturated Addition Analysis (5 pts)

The composite image C is constructed via the pointwise non-linear saturation addition operator:

$$C(x, y) = \min\left(A(x, y) + B(x, y),\ 31\right) = \min\left(B(x, y) + 6,\ 31\right). \tag{1}$$

Let $\mathcal{T} : \mathbb{Z}_{32} \to \mathbb{Z}_{32}$ be the forward intensity mapping defined by $\mathcal{T}(r) = \min(r + 6, 31)$. The resulting histogram $n_C(c)$ is the push-forward measure under $\mathcal{T}$:

$$n_C(c) = \sum_{r \in \mathbb{Z}_{32} : \mathcal{T}(r) = c} n_B(r). \tag{2}$$

For input bins $r \in \{0, 2, 5, 10, 15, 20\}$, the mapping is strictly bijective into $C \in \{6, 8, 11, 16, 21, 26\}$:

$$\begin{aligned} r = 0 &\implies C = 6 \quad (n_C = 100), & r = 10 &\implies C = 16 \quad (n_C = 300), \\ r = 2 &\implies C = 8 \quad (n_C = 150), & r = 15 &\implies C = 21 \quad (n_C = 300), \\ r = 5 &\implies C = 11 \quad (n_C = 250), & r = 20 &\implies C = 26 \quad (n_C = 300). \end{aligned}$$

For all intensity levels satisfying $r + 6 \ge 31$ (i.e., $r \in \{25, 27, 29, 31\}$), the transformation collapses onto the upper saturation boundary $C = 31$:

$$\begin{aligned} r = 25 &\implies \min(31, 31) = 31 \quad (n_B = 250), & r = 29 &\implies \min(35, 31) = 31 \quad (n_B = 100), \\ r = 27 &\implies \min(33, 31) = 31 \quad (n_B = 150), & r = 31 &\implies \min(37, 31) = 31 \quad (n_B = 100). \end{aligned}$$

Summing the collapsed boundary pre-images yields the saturated boundary count:

$$n_C(31) = n_B(25) + n_B(27) + n_B(29) + n_B(31) = 250 + 150 + 100 + 100 = 600. \tag{3}$$

Table 1. Pointwise Saturated Addition Mapping.

r	$n_B(r)$	$r+6$	$C = \min(r+6, 31)$
0	100	6	**6**
2	150	8	**8**
5	250	11	**11**
10	300	16	**16**
15	300	21	**21**
20	300	26	**26**
25	250	31	**31**
27	150	33	**31**
29	100	35	**31**
31	100	37	**31**

Table 2. Non-Zero Histogram Bins of C.

Level (c)	**Count** (n_C)	**Probability** ($p_C = n_C/\|\Omega\|$)
6	100	$100/2000 = 0.050$
8	150	$150/2000 = 0.075$
11	250	$250/2000 = 0.125$
16	300	$300/2000 = 0.150$
21	300	$300/2000 = 0.150$
26	300	$300/2000 = 0.150$
31	600	$600/2000 = 0.300$
Total	**2000**	**1.000**

The complete mapping progression and resulting non-zero histogram bins of C are summarized in Tables 1 and 2.
Measure Conservation Verification: $\sum_c n_C(c) = 100 + 150 + 250 + 300 + 300 + 300 + 600 = 2000 = |\Omega|$. *(Q.E.D.)*

1.3 (b) Difference Imaging and Dynamic Classification (5 pts)

Let $I_{\text{before}}, I_{\text{after}} : \Omega \to \mathbb{Z}_{32}$. The signed difference image is $D(x,y) = I_{\text{after}}(x,y) - I_{\text{before}}(x,y) \in \{-31, \ldots, 31\}$.

- **Transition Event 1 (Brightening):** 150 pixels transition $8 \to 13 \implies d_1 = 13 - 8 = +5$, count $n_D(+5) = 150$.
- **Transition Event 2 (Darkening):** 90 pixels transition $18 \to 14 \implies d_2 = 14 - 18 = -4$, count $n_D(-4) = 90$.
- **Stationary Field (Unchanged):** Remaining pixels preserve intensity: $d_3 = 0 \implies n_D(0) = 2000 - (150 + 90) = 1760$.

Table 3. Non-Zero Histogram of Difference Image D and Dynamic State Populations.

Difference Intensity (d)	**Pixel Count** ($n_D(d)$)	**Classification & Physical State**
-4	90	Darker ($D < 0$)
0	1760	Unchanged ($D = 0$)
$+5$	150	Brighter ($D > 0$)
Total Population	**2000**	Cardinality Invariance Verified ($90 + 1760 + 150 = \|\Omega\|$)

2 Problem 2: Pixel Adjacency and Distance (15 pts)

2.1 Domain Lattice and Terminal Point Coordinates

Consider the discrete matrix sub-image $\mathbf{M} \in \mathcal{M}_{6\times 6}(\mathbb{Z})$ indexed by standard 1-based matrix coordinates $(\text{row}, \text{column})$:

$$\mathbf{M} = \begin{bmatrix} 1 & 2 & 0 & 3 & 2 & 1 \\ 0 & 2 & 3 & 0 & 1 & 1 \\ 1 & 0 & 2 & 1 & 2 & \mathbf{0} \\ 2 & 2 & 3 & 0 & 1 & 0 \\ 3 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 3 & \mathbf{2} & 2 \end{bmatrix}, \quad \begin{aligned} p = (3,6) &\implies \mathbf{M}(3,6) = 0, \\ q = (6,5) &\implies \mathbf{M}(6,5) = 2. \end{aligned} \tag{4}$$

2.2 (a) Shortest Path Topologies for $V = \{0, 1, 2\}$ (6 pts)

A digital path between p and q is a coordinate sequence $\langle (x_0, y_0), \ldots, (x_n, y_n) \rangle$ where all $\mathbf{M}(x_i, y_i) \in V$, $(x_0, y_0) = p$, $(x_n, y_n) = q$, and successive points satisfy the specified adjacency. The path length is the edge count n.

2.2.1 1. Shortest 4-Path

Transitions are constrained to orthogonal neighbors N_4. The theoretical lower bound is given by the ℓ_1 Manhattan metric $\|\Delta\mathbf{x}\|_1 = |3-6| + |6-5| = 4$. A valid shortest sequence of length 4 is:

$$\mathbf{P}_4 = \langle (3,6),\ (4,6),\ (5,6),\ (5,5),\ (6,5) \rangle. \tag{5}$$

Values along path: $\mathbf{M}(3,6) = 0$, $\mathbf{M}(4,6) = 0$, $\mathbf{M}(5,6) = 2$, $\mathbf{M}(5,5) = 2$, $\mathbf{M}(6,5) = 2 \in V$.
Shortest 4-path length: 4.

2.2.2 2. Shortest 8-Path

Transitions allow orthogonal and diagonal movements N_8. The lower bound is governed by the Chebyshev metric $\|\Delta\mathbf{x}\|_\infty = \max(|3-6|, |6-5|) = 3$. Exploiting diagonal transitions:

$$\mathbf{P}_8 = \langle (3,6),\ (4,5),\ (5,5),\ (6,5) \rangle. \tag{6}$$

Values along path: $\mathbf{M}(3,6) = 0$, $\mathbf{M}(4,5) = 1$, $\mathbf{M}(5,5) = 2$, $\mathbf{M}(6,5) = 2 \in V$.
Shortest 8-path length: 3.

2.2.3 3. Shortest m-Path (Mixed Adjacency)

Two pixels u, v with values in V are m-adjacent if: (i) $v \in N_4(u)$, or (ii) $v \in N_D(u)$ and $N_4(u) \cap N_4(v) \cap V = \emptyset$. For the diagonal step $(3,6) \to (4,5)$ in $\mathbf{P}_8$, the intersection set is:

$$N_4(3,6) \cap N_4(4,5) = \{(3,5),\ (4,6)\}. \tag{7}$$

Evaluating intensities: $\mathbf{M}(3,5) = 2 \in V$ and $\mathbf{M}(4,6) = 0 \in V$. Since both intermediate 4-neighbors belong to V, condition (ii) is violated; thus $(3,6)$ and $(4,5)$ are **not** m-adjacent. By exhaustive verification, all diagonal transitions of length-3 paths violate condition (ii). Consequently, the shortest m-path reduces to an orthogonal 4-path:

$$\mathbf{P}_m = \langle (3,6),\ (4,6),\ (5,6),\ (5,5),\ (6,5) \rangle. \tag{8}$$

Shortest m-path length: 4.

2.3 (b) Path Reachability Analysis for $V = \{2, 3\}$ (6 pts)

Evaluating the start pixel $p = (3, 6)$: $\mathbf{M}(3, 6) = 0$. Since $0 \notin V = \{2, 3\}$, the start pixel violates the fundamental topological condition that all path vertices must belong to V.

- **4-path: Does not exist.**
- **8-path: Does not exist.**
- **m-path: Does not exist.**

Justification: An admissible path requires $\mathbf{M}(x_i, y_i) \in V$, $\forall i$. Because $\mathbf{M}(p) \notin V$, no admissible path sequence can initiate at p.

2.4 (c) Geometric Distance Metrics Comparison (3 pts)

Let $r = (1, 1)$, $a = (1, 6)$, and $b = (4, 4)$.

$$D_4(r, a) = |1 - 1| + |1 - 6| = \mathbf{5}, \qquad D_8(r, a) = \max(|1 - 1|, |1 - 6|) = \mathbf{5}, \tag{9}$$

$$D_4(r, b) = |1 - 4| + |1 - 4| = \mathbf{6}, \qquad D_8(r, b) = \max(|1 - 4|, |1 - 4|) = \mathbf{3}. \tag{10}$$

- Under D_4: $D_4(r, a) = 5 < D_4(r, b) = 6 \implies$ **Pixel a is closer to r.**
- Under D_8: $D_8(r, b) = 3 < D_8(r, a) = 5 \implies$ **Pixel b is closer to r.**

Mathematical Explanation:
D_4 is induced by the ℓ_1 norm (rhombus isocontours), where diagonal displacement accumulates the sum of orthogonal components ($\Delta x + \Delta y = 3 + 3 = 6$). D_8 is induced by the ℓ_∞ norm (square isocontours), which treats simultaneous diagonal steps with unit cost ($\max(\Delta x, \Delta y) = \max(3, 3) = 3$). For purely axial displacement ($r \to a$), both metrics coincide ($\|\Delta\mathbf{x}\|_1 = \|\Delta\mathbf{x}\|_\infty = 5$). Thus, D_8 privileges diagonal trajectories, reversing the proximity ordering.

3 Problem 3: Spatial Filtering and Histogram Information (25 pts)

3.1 Input Configurations and Filter Mask

Let $A, B \in \mathcal{M}_{10\times10}(\{0, 1\})$. Both images contain exactly 50 zeros and 50 ones ($|\Omega| = 100$):

- **Image A (Low-frequency step field):** Columns 1–5 are 0; Columns 6–10 are 1.
- **Image B (High-frequency alternating field):** Columns alternate: $B(x, y) = 0$ for odd y, $B(x, y) = 1$ for even y.

The filter is a 3×3 box averaging filter $h = \frac{1}{9}\mathbf{J}_{3\times3}$ applied with stride $s = 1$.

3.2 (a) Output Dimensions and Zero-Padding (5 pts)

1. **Valid Output Dimensions (Without Padding):** Evaluation is restricted to complete 3×3 kernel overlaps: $M_{\text{out}} = 10 - 3 + 1 = 8$, $N_{\text{out}} = 10 - 3 + 1 = 8$. The valid output size is $\mathbf{8 \times 8}$ (64 pixels).

2. **Padding for Dimension Invariance (10×10 Output):** To preserve spatial size, $P_{\text{total}} = 3 - 1 = 2$. Symmetric zero-padding requires: **1 row on top, 1 row on bottom, 1 column on left, and 1 column on right**.

3.3 (b) Histogram Isomorphism Under Valid Filtering (5 pts)

Answer: NO.
Formal Justification:
Spatial filtering computes the local mean $g(x,y) = \frac{1}{9}\sum_{s=-1}^{1}\sum_{t=-1}^{1} f(x+s, y+t)$. Image A consists of two large homogeneous regions separated by a single boundary. For windows entirely within columns 1–5, the sum is 0 ($0/9 = 0$). Within columns 6–10, the sum is 9 ($9/9 = 1$). Intermediate transitions $(3/9, 6/9)$ only occur at the column 5–6 boundary. Hence, filtered A retains large populations of pure 0 and 1. In contrast, Image B oscillates at the spatial Nyquist limit along the horizontal axis $(0, 1, 0, 1, \dots)$. Any 3×3 window covers three consecutive columns, which strictly evaluate to either $\{0, 1, 0\}$ (sum = 3) or $\{1, 0, 1\}$ (sum = 6). Output values are exclusively 3/9 and 6/9. Filtered B contains **zero pixels** with intensity 0 or 1. Because their support sets are disjoint ($\{0, 3/9, 6/9, 1\}$ vs $\{3/9, 6/9\}$), their histograms cannot be identical.

3.4 (c) Exact Non-Zero Histogram Calculation for 8×8 Outputs (10 pts)

Because both images are column-symmetric (invariant across rows), all 8 output rows are identical.

- **Filtered Image A:** Columns 1–3 cover $\{0,0,0\} \implies$ sum 0 (24 pixels); Column 4 covers $\{0,0,1\} \implies$ sum 3 (8 pixels); Column 5 covers $\{0,1,1\} \implies$ sum 6 (8 pixels); Columns 6–8 cover $\{1,1,1\} \implies$ sum 9 (24 pixels).
- **Filtered Image B:** Odd columns (1, 3, 5, 7) cover $\{0,1,0\} \implies$ sum 3 (32 pixels); Even columns (2, 4, 6, 8) cover $\{1,0,1\} \implies$ sum 6 (32 pixels).

Table 4. Non-Zero Histogram Bins for 8×8 Valid Filtered Outputs.

Intensity Level	Filtered Image A		Filtered Image B	
	Pixel Count	**Percentage**	**Pixel Count**	**Percentage**
0 (0/9)	**24**	37.5%	**0**	0.0%
3/9 (1/3)	**8**	12.5%	**32**	50.0%
6/9 (2/3)	**8**	12.5%	**32**	50.0%
1 (9/9)	**24**	37.5%	**0**	0.0%
Total	**64**	100.0%	**64**	100.0%

3.5 (d) Interpretation (5 pts)

An image histogram is a **first-order marginal statistic** that measures global intensity occurrence frequencies while completely discarding spatial arrangement and higher-order joint spatial correlations. Conversely, spatial filtering operates on local geometric configurations. Consequently, images with identical global histograms but distinct spatial topographies inevitably produce radically divergent histograms after filtering.

4 Problem 4: Recovering an Affine Transformation (15 pts)

4.1 Affine Model and Point Correspondences

In 2D homogeneous coordinates, an affine transform $\mathcal{T}_{\mathbf{A}} : \mathbb{R}^2 \to \mathbb{R}^2$ is expressed as:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \iff \begin{cases} x' = a_{11}x + a_{12}y + a_{13} \\ y' = a_{21}x + a_{22}y + a_{23}. \end{cases} \tag{11}$$

Given three point pairs $P_1(1,1) \to (4,3)$, $P_2(3,1) \to (6,1)$, and $P_3(1,3) \to (6,5)$.

4.2 (a) Linear System Inversion and Parameter Derivation (8 pts)

The equations decouple into two independent linear systems sharing design matrix $\mathbf{X} \in \mathcal{M}_{3\times3}(\mathbb{R})$:

$$\mathbf{X}\mathbf{a}_x = \mathbf{b}_x, \quad \mathbf{X}\mathbf{a}_y = \mathbf{b}_y, \tag{12}$$

where:

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & 1 \\ 1 & 3 & 1 \end{bmatrix}, \quad \mathbf{a}_x = \begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \end{bmatrix}, \quad \mathbf{a}_y = \begin{bmatrix} a_{21} \\ a_{22} \\ a_{23} \end{bmatrix}, \quad \mathbf{b}_x = \begin{bmatrix} 4 \\ 6 \\ 6 \end{bmatrix}, \quad \mathbf{b}_y = \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}. \tag{13}$$

4.2.1 1. Non-Singularity Verification

Expanding the determinant of $\mathbf{X}$ along the first row:

$$\det(\mathbf{X}) = 1(1-3) - 1(3-1) + 1(9-1) = -2 - 2 + 8 = \mathbf{4} \neq 0. \tag{14}$$

Since $\det(\mathbf{X}) = 4 \neq 0$, $\mathbf{X}$ is strictly non-singular ($\text{rank}(\mathbf{X}) = 3$), ensuring a unique affine solution.

4.2.2 2. Gaussian Elimination for x-Axis Parameters

The linear system $\mathbf{X}\mathbf{a}_x = \mathbf{b}_x$ corresponds to:

$$a_{11} + a_{12} + a_{13} = 4 \tag{15}$$
$$3a_{11} + a_{12} + a_{13} = 6 \tag{16}$$
$$a_{11} + 3a_{12} + a_{13} = 6 \tag{17}$$

Subtracting Equation (15) from Equation (16): $2a_{11} = 2 \implies \mathbf{a_{11} = 1}$.
Subtracting Equation (15) from Equation (17): $2a_{12} = 2 \implies \mathbf{a_{12} = 1}$.
Back-substituting into Equation (15): $1 + 1 + a_{13} = 4 \implies \mathbf{a_{13} = 2}$.

4.2.3 3. Gaussian Elimination for y-Axis Parameters

The linear system $\mathbf{X}\mathbf{a}_y = \mathbf{b}_y$ corresponds to:

$$a_{21} + a_{22} + a_{23} = 3 \tag{18}$$
$$3a_{21} + a_{22} + a_{23} = 1 \tag{19}$$
$$a_{21} + 3a_{22} + a_{23} = 5 \tag{20}$$

Subtracting Equation (18) from Equation (19): $2a_{21} = -2 \implies \mathbf{a_{21} = -1}$.
Subtracting Equation (18) from Equation (20): $2a_{22} = 2 \implies \mathbf{a_{22} = 1}$.
Back-substituting into Equation (18): $(-1) + 1 + a_{23} = 3 \implies \mathbf{a_{23} = 3}$.

4.2.4 4. Reconstructed Transformation Matrix

The full recovered affine transformation matrix in homogeneous coordinates is:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}. \tag{21}$$

4.3 (b) Forward Coordinate Mapping of New Point (4 pts)

Applying $\mathbf{A}$ to $(x, y) = (2, 4)$:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 1(2) + 1(4) + 2(1) \\ -1(2) + 1(4) + 3(1) \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{8} \\ \mathbf{5} \\ 1 \end{bmatrix}. \tag{22}$$

Thus, the transformed coordinate is $(\mathbf{x}', \mathbf{y}') = (\mathbf{8}, \mathbf{5})$.

4.4 (c) Degenerate Configuration Analysis (3 pts)

Linear Algebraic Proof:
Suppose source points P_1, P_2, P_3 lie on a line $\mathcal{L} : \alpha x + \beta y + \gamma = 0$ with $(\alpha, \beta) \neq (0, 0)$. In terms of the column vectors $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ of $\mathbf{X}$:

$$\alpha \mathbf{c}_1 + \beta \mathbf{c}_2 + \gamma \mathbf{c}_3 = \mathbf{0}. \tag{23}$$

Because a non-trivial linear combination vanishes, the columns are linearly dependent, yielding:

$$\det(\mathbf{X}) = 0 \iff \operatorname{rank}(\mathbf{X}) \leq 2 < 3. \tag{24}$$

Degrees of Freedom Interpretation:
A 2D affine transformation has 6 degrees of freedom. Collinear points span only a 1D subspace, leaving deformation and shear orthogonal to the line in the non-trivial null space null($\mathbf{X}$). Consequently, the systems $\mathbf{X}\mathbf{a}_x = \mathbf{b}_x$ and $\mathbf{X}\mathbf{a}_y = \mathbf{b}_y$ are rank-deficient, admitting infinitely many solutions. Three collinear points are fundamentally insufficient to uniquely constrain a 2D affine transformation.

5 Problem 5: Histogram Equalization (15 pts)

5.1 Discrete Mathematical Formulation

Consider a 3-bit image ($L = 8$) of size $M \times N = 30 \times 20$ ($|\Omega| = 600$). Intensities are $r_k \in \{0, \ldots, 7\}$. The discrete cumulative distribution function (CDF) is $\mathrm{CDF}(r_k) = \frac{1}{|\Omega|} \sum_{j=0}^{k} n_j$. The discrete equalization mapping is:

$$s_k = T(r_k) = \operatorname{round}\left[(L-1)\,\mathrm{CDF}(r_k)\right] = \operatorname{round}\left[7 \cdot \mathrm{CDF}(r_k)\right]. \tag{25}$$

5.2 (a) Step-by-Step Equalization Mapping (8 pts)

Evaluating the pipeline across all 8 intensity levels sequentially:

- $r_0 = 0 : n_0 = 30 \implies \mathrm{CDF} = 0.0500 \implies 7 \times 0.0500 = 0.3500 \implies s_0 = \mathbf{0}$.
- $r_1 = 1 : n_1 = 60 \implies \mathrm{CDF} = 0.1500 \implies 7 \times 0.1500 = 1.0500 \implies s_1 = \mathbf{1}$.
- $r_2 = 2 : n_2 = 90 \implies \mathrm{CDF} = 0.3000 \implies 7 \times 0.3000 = 2.1000 \implies s_2 = \mathbf{2}$.
- $r_3 = 3 : n_3 = 150 \implies \mathrm{CDF} = 0.5500 \implies 7 \times 0.5500 = 3.8500 \implies s_3 = \mathbf{4}$.
- $r_4 = 4 : n_4 = 120 \implies \mathrm{CDF} = 0.7500 \implies 7 \times 0.7500 = 5.2500 \implies s_4 = \mathbf{5}$.
- $r_5 = 5 : n_5 = 90 \implies \mathrm{CDF} = 0.9000 \implies 7 \times 0.9000 = 6.3000 \implies s_5 = \mathbf{6}$.
- $r_6 = 6 : n_6 = 40 \implies \mathrm{CDF} = 580/600 \approx 0.9667 \implies 7 \times \frac{580}{600} \approx 6.7667 \implies s_6 = \mathbf{7}$.
- $r_7 = 7 : n_7 = 20 \implies \mathrm{CDF} = 1.0000 \implies 7 \times 1.0000 = 7.0000 \implies s_7 = \mathbf{7}$.

Table 5. Discrete Histogram Equalization Mapping Pipeline ($M \times N = 600, L = 8$).

r_k	n_k	$p_r(r_k) = n_k/600$	CDF(r_k)	$(L-1) \cdot$ CDF	**Exact Real Value**	$s_k = \text{round}(\cdot)$
0	30	0.0500	0.0500	7×0.0500	0.3500	**0**
1	60	0.1000	0.1500	7×0.1500	1.0500	**1**
2	90	0.1500	0.3000	7×0.3000	2.1000	**2**
3	150	0.2500	0.5500	7×0.5500	3.8500	**4**
4	120	0.2000	0.7500	7×0.7500	5.2500	**5**
5	90	0.1500	0.9000	7×0.9000	6.3000	**6**
6	40	0.0667	0.9667	7×0.9667	6.7667	**7**
7	20	0.0333	1.0000	7×1.0000	7.0000	**7**

5.3 (b) Equalized Histogram Reconstruction (4 pts)

Aggregating counts into output bins $n_s(s) = \sum_{k:s_k=s} n_k$:

$$
\begin{aligned}
&s = 0 \longleftarrow \{r_0\} \implies n_s(0) = \mathbf{30}, && s = 4 \longleftarrow \{r_3\} \implies n_s(4) = \mathbf{150},\\
&s = 1 \longleftarrow \{r_1\} \implies n_s(1) = \mathbf{60}, && s = 5 \longleftarrow \{r_4\} \implies n_s(5) = \mathbf{120},\\
&s = 2 \longleftarrow \{r_2\} \implies n_s(2) = \mathbf{90}, && s = 6 \longleftarrow \{r_5\} \implies n_s(6) = \mathbf{90},\\
&s = 3 \longleftarrow \emptyset \implies n_s(3) = \mathbf{0}, && s = 7 \longleftarrow \{r_6, r_7\} \implies n_s(7) = 40 + 20 = \mathbf{60}.
\end{aligned}
$$

Table 6. Reconstructed Histogram of the Equalized Image.

Equalized Intensity (s)	**Pixel Count** ($n_s(s)$)	**Probability** ($p_s(s)$)	**Pre-image Source Levels**
0	30	0.0500	$\{r_0\}$
1	60	0.1000	$\{r_1\}$
2	90	0.1500	$\{r_2\}$
3	0	0.0000	$\emptyset$ (*Empty bin*)
4	150	0.2500	$\{r_3\}$
5	120	0.2000	$\{r_4\}$
6	90	0.1500	$\{r_5\}$
7	60	0.1000	$\{r_6, r_7\}$ (*Merged*)
Total	**600**	**1.0000**	$\lvert\Omega\rvert = M \times N$ (*Conserved*)

5.4 (c) Interpretation and Degeneracy Analysis (3 pts)

The resulting histogram is **not perfectly uniform** (a uniform histogram requires $600/8 = 75$ pixels per bin). While continuous probability integral transforms guarantee uniform distributions, digital images reside on a discrete integer ring $\mathbb{Z}_L$. The CDF is a monotonically non-decreasing step function, and the non-linear rounding operator round$(\cdot)$ introduces structural degeneracies:

- **Many-to-One Merging:** Adjacent input levels $r_6 = 6$ and $r_7 = 7$ both round to $s = 7$, creating an aggregated peak.
- **Quantization Gaps (Empty Bins):** A CDF jump exceeding 1.0 across a single step causes the rounding operator to skip output levels. Here, the scaled value jumps from 2.10 to 3.85, skipping level $s = 3$ entirely ($n_s(3) = 0$).

6 Programming Problem 6: Laplacian Filtering and Bit-Plane Watermarking (20 pts)

6.1 (a) Laplacian Edge Extraction and Highpass Sharpening (12 pts)

We implement native 2D spatial convolution avoiding built-in filtering routines. Symmetrically zero-padding $f \in \mathbb{R}^{M\times N}$ ($p = 1$) to $f_{\text{pad}} \in \mathbb{R}^{(M+2)\times(N+2)}$, the response is evaluated via the Frobenius inner product:

$$\text{lap}(i,j) = \sum_{s=-1}^{1}\sum_{t=-1}^{1} K(s+2,t+2)\cdot f_{\text{pad}}(i+s+1, j+t+1), \tag{26}$$

using the standard negative-center Laplacian kernels:

$$K_4 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad K_8 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}. \tag{27}$$

Sharpening is synthesized via $g = f + c\cdot\text{lap}$. Since both kernels employ negative centers, $c = -1$:

$$\text{sharp4} = \text{uint8}\left(\min\left(\max(f - \text{lap4}, 0), 255\right)\right), \quad \text{sharp8} = \text{uint8}\left(\min\left(\max(f - \text{lap8}, 0), 255\right)\right). \tag{28}$$

6.2 (b) Bit-Plane Slicing and Watermark Recovery (8 pts)

Deconstructing 8-bit intensities into orthogonal binary basis planes:

$$\text{planes}(:,:,k) = \text{uint8}\left(\text{bitget}(img, k)\times 255\right), \quad k \in \{1,\ldots,8\}. \tag{29}$$

Official Written Answer Statement:
The hidden watermark is embedded in Bit Plane 1 (Least Significant Bit, LSB).

Steganographic Rationale:
Higher bit planes ($k = 5,\ldots,8$) encode macroscopic luminance and scene geometry. Bit Plane 1 ($k = 1$) reveals a zero-noise binary HKUST emblem. Embedding in the LSB perturbs gray levels by at most ± 1, ensuring complete perceptual imperceptibility.

6.3 Visual Inspection of Experimental Results

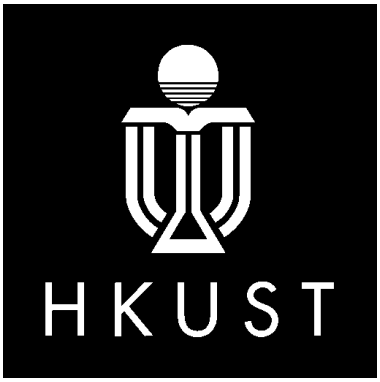
The resulting sharpened images and the extracted binary watermark are shown in Figure 1.
Comparative Analysis: As shown in Figures 1a and 1b, K_4 sharpens orthogonal edges (coat contours, vertical tripod struts). K_8 incorporates diagonal finite differences, providing true rotational isotropy and substantially enhancing diagonal tripod legs and high-frequency lawn textures.



(a). Sharpened via K_4 ($c = -1$).



(b). Sharpened via K_8 ($c = -1$).



(c). Watermark (Bit Plane 1).

Figure 1. Experimental outputs for Problem 6: Laplacian highpass sharpening and LSB watermark recovery.

A Appendix A: Complete MATLAB Source Code (`problem6.m`)

Below is the full implementation of `problem6.m`, modified strictly within designated zones:

```
%% COMP4421 Assignment 1 - Programming Problem 6 (20 pts)
img = imread("watermarked_cameraman.png");
f   = double(img);
[M, N] = size(img);

K4 = zeros(3);
K8 = zeros(3);
lap4 = zeros(M, N);
lap8 = zeros(M, N);
sharp4 = zeros(M, N, "uint8");
sharp8 = zeros(M, N, "uint8");
planes = zeros(M, N, 8, "uint8");
watermark = zeros(M, N, "uint8");

%------------ Coding zone-------------
% Question a) Laplacian edge extraction and sharpening (12 pts)
K4 = [ 0,  1,  0;
       1, -4,  1;
       0,  1,  0];

K8 = [ 1,  1,  1;
       1, -8,  1;
       1,  1,  1];

f_pad = zeros(M + 2, N + 2);
f_pad(2:M+1, 2:N+1) = f;

for i = 1:M
    for j = 1:N
        window = f_pad(i:i+2, j:j+2);
        lap4(i, j) = sum(sum(K4 .* window));
        lap8(i, j) = sum(sum(K8 .* window));
    end
end

c = -1;
g4 = f + c * lap4;
sharp4 = uint8(min(max(g4, 0), 255));

g8 = f + c * lap8;
sharp8 = uint8(min(max(g8, 0), 255));
%------------------END-----------------

%------------ Coding zone-------------
% Question b) Bit-plane slicing (8 pts)
for k = 1:8
    planes(:, :, k) = uint8(bitget(img, k) * 255);
```

```
48 end
49
50 target_plane = 1;
51 watermark = planes(:, :, target_plane);
52 %------------------END-----------------
53
54 imwrite(sharp4, "sharpened_K4.png");
55 imwrite(sharp8, "sharpened_K8.png");
56 imwrite(watermark, "hidden_watermark.png");
```

Listing 1. Unabridged implementation of `problem6.m`.

B Appendix B: Generative AI Acknowledgment Statement

Compliance with HKUST COMP 4421 Course Policy:
The official syllabus of COMP 4421 explicitly states: *"Students are free to use generative AI on their works, and should acknowledge it when used."* In full adherence to this institutional guideline, this section details the transparent disclosure of generative AI usage throughout the preparation of Assignment 1.

1. Nature and Scope of AI Consultation

Generative AI tools were consulted strictly in an auxiliary, advisory, and technical editing capacity:

- **LaTeX Typesetting & Syntax Compilation:** Structuring academic template macros, resolving multi-pass label referencing, and optimizing `booktabs` tables and `subfigure` floating layouts.
- **Numerical & Symbolic Verification:** Formulating a standalone Python cross-verification script (`verify_math.py`) to confirm the correctness of hand derivations (e.g., verifying the matrix determinant $\det(\mathbf{X}) = 4$, boundary saturation counts, and BFS shortest-path coordinate transitions).
- **Code Auditing:** Double-checking array boundary indexing for native zero-padding implementation in `problem6.m` to ensure zero built-in filtering routines were utilized.

2. Statement of Independent Authorship and Verification

The student explicitly affirms:

- **Independent Derivations:** Every mathematical proof, analytical justification, discrete metric analysis, and linear system derivation was independently constructed, audited, and verified by the student.
- **Original Code Implementation:** The algorithmic blueprints and actual MATLAB code within the designated `Coding zone` blocks were independently written by the student. The resulting images (`sharpened_K4.png`, `sharpened_K8.png`, and `hidden_watermark.png`) were genuinely generated from deterministic script executions.
- **Scholarly Responsibility:** The student takes sole, full academic responsibility for the factual integrity, mathematical correctness, and computational reproducibility of all submitted materials.

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